KGML gasmet data processing

CO2 and N2O

Jesse Bealsburg

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Key points

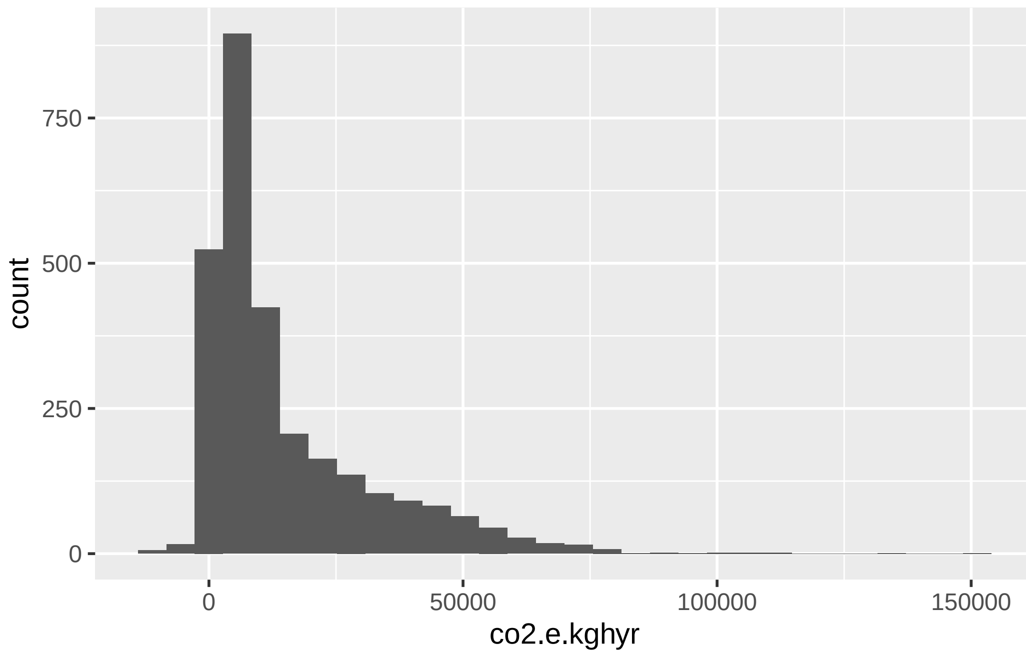
- Converted ppm of CO2 to kg C ha yr and N2O to kg N ha yr
  - fitting lines, extracting slope
- Most CO2 fluxes are positive, which is what it should be.
  - 105 of the 2836 predict positive CO2 emissions
- ~50% of N2O fluxes are positive.

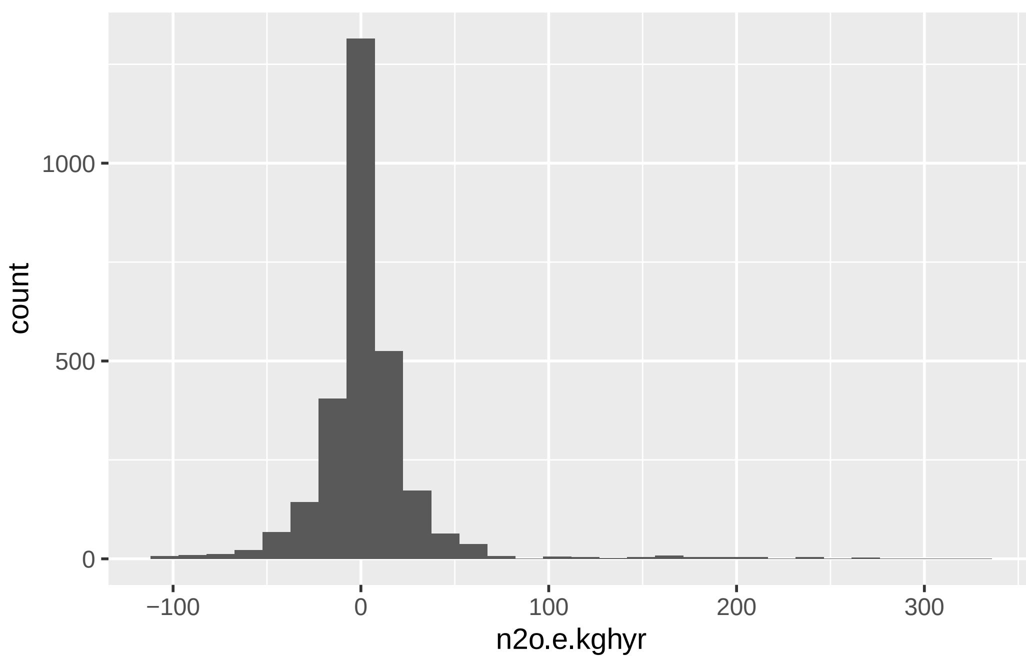
Yufeng wants N2O data converted from the ppm unit to an amount of N released over an area and amount of time. Here I convert N2O ppm to kg N ha over a year by modifying the script used by Jake Kundert to convert CO2 to kg C ha yr.

I am importing the rmarkdown file authored by Jake Kundert that I modified, exporting it as a R script and running that rscript to populate my environment, then removing most of the objects in the environment except the ones I am keeping

This data has already undergone some QAQC. I will simply join and export here. In general, CO2 data is a good indicator of the quality of the N2O data. If the standard error was abnormally high for CO2, that same sample likely has unreliable N2O data.

Visualization






```
# A tibble: 149 x 5
```

```
# Groups:   experiment [5]
   experiment date       totalSlopeTally negSlopeTally negFluxPercent
   <fct>      <chr>                <int>         <int>          <dbl>
 1 atat       2023-06-01              10             8           80  
 2 ncas       2024-10-08              28            20           71.4
 3 ncas       2024-09-24              27            19           70.4
 4 eddyflux   2023-05-24              10             7           70  
 5 ncas       2023-06-05              36            25           69.4
 6 ncas       2024-04-09              16            11           68.8
 7 eddyflux   2022-06-28              21            14           66.7
 8 eddyflux   2024-07-29              12             8           66.7
 9 ncas       2024-03-19              14             9           64.3
10 ncas       2024-04-25              14             9           64.3
# i 139 more rows
```